

Semester: Jan 2025 - April 2025		
Maximum Marks: 30	Examination: Sample In-Semester Examination	tion :1 hr.15 min.
Programme code: 03 Programme: B.Tech Electronics and Telecommunication Engineering	Class: SY	Semester: IV (SVU 2023)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: EXTC	
Course Code: 216U03C401	Name of the Course: Probability, Statistics and Optimization Techniques	

Question No.		Max. Marks
Q1a	The following stem and leaf plot gives the scores on a statistics exam taken by industrial engineering students. <pre> 9 0, 1, 4 8 3, 5, 5, 7, 8 7 2, 4, 4, 5, 7, 7, 8 6 0, 2, 3, 4, 6, 6 5 2, 5, 5, 6, 8 4 3, 6 </pre> Find 1. the sample mean 2. the sample median 3. the sample mode 4. the sample variance and SD OR The total number of hours, measured in units of 1 hour, that a family runs a vacuum cleaner over a period of one year is a continuous random variable X that has the density function $f(x) = \begin{cases} x, & 0 < x < 1 \\ (2-x), & 1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$ Find the probability that over a period of one year, a family runs their vacuum cleaner (a) less than 1.2 hours; (b) between 0.5 and 1 hour	05

Q1b	A spam filter uses Bayes' theorem to classify emails as spam or not spam. 40% of all received emails are spam. 5% of spam emails contain the word "lottery". 1% of non-spam emails also contain the word "lottery". If an email contains the word "lottery," what is the probability that it is spam?	05
Q2	A university conducts a standardized test, and the test scores are normally distributed with: • Mean (μ) = 75 marks • Standard deviation (σ)= 12 marks • A professor selects a random sample of 36 students. Find the probability that the sample mean score of the 36 students is greater than 78 marks. OR Derive maximum-likelihood estimators for parameter λ based on Poisson random variable	10
Q3	Suppose an IT company has two stores that sell computers. The company recorded the number of sales each store made each month. In the past 12 months, we have the following numbers of sold computers: Store 1: 350, 460, 20, 160, 580, 250, 210, 120, 200, 510, 290, 380. Store 2: 520, 180, 260, 380, 80, 500, 630, 420, 210, 70, 440, 140. Draw the box plot for both the companies. Using Box plot find which store performs better. Also write the interpretations, outliers and skewness drawn from the plots.	10